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FIFTH SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

REINFORCEMENT LEARNING

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Relate why the target policy in Q-learning greedy for the current action-value estimates?	L2	CO4	**02**
	b.	Explain between Sarsa and expected Sarsa, which algorithm's update requires more computation per step?	L1	CO4	**02**
	c.	Describe k-armed bandit problem with respect to reinforcement learning.	L1	CO1	**02**
	d.	Illustrate the action-values Q function.	L1	CO1	**02**
	e.	Define the epsilon-greedy method.	L1	CO2	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Summarize the selection of actions greedily using an action-value function.	L2	CO2	**05**
		OR			
	b.	Compare the short-term benefits of exploitation and the long-term benefits of exploration.	L2	CO1	**05**
3.	**a.**	Describe the upper confidence bound action selection method.	L1	CO1	**05**
		OR			
	b.	Illustrate the action-value estimation methods and explain their usage in one reinforcement learning technique.	L1	CO1	**05**
4.	**a.**	Frame the Markov Decision Processes (MDP) working using a graphical representation of any fully visible environment.	L5	CO2	**05**
		OR			
	b.	Describe how rewards relate to the goal of an agent.	L4	CO1	**05**
5.	**a.**	Connect, when to formalize a task as episodic or continuing.	L4	CO3	**05**
		OR			
	b.	Compare the similarities and differences between stochastic and deterministic policies	L2	CO6	**05**
6.	**a.**	Interpret and derive the Bellman equation for state-value functions.	L2	CO3	**05**
		OR			
	b.	Correlate how the Bellman optimality equations relate to the Bellman equations.	L4	CO2	**05**

7.	**a.**	Explain the setting in which dynamic programming can be applied, as well as explain its limitations.	L1	CO2	**05**
		OR			
	b.	Outline the policy iteration algorithm for finding the optimal policy.	L3	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Demonstrate how Monte-Carlo methods can be used to estimate value functions from sampled interaction.	L2	CO3	**10**
		OR			
	b.	Demonstrate how off-policy learning can help deal with the exploration problem	L2	CO5	**10**
9.	**a.**	Define temporal-difference learning and temporal-difference (TD) error.	L1	CO4	**10**
		OR			
	b.	Identify key advantages of TD methods over Dynamic Programming and Monte Carlo methods	L3	CO5	**10**
10.	**a.**	Simulate how both direct RL and planning updates can be combined through the Dyna architecture?	L6	CO5	**10**
		OR			
	b.	Defend how Dyna can plan successfully with a partially inaccurate model.	L5	CO5	**10**
11.	**a.**	Weigh the compactness of the sample model representation with the distribution models.	L5	CO4	**10**
		OR			
	b.	Defend how planning is used to improve policies	L5	CO6	**10**
12.	**a.**	Describe the Sarsa Control algorithm. Explain how generalized policy iteration can support TD to find improved policies.	L5	CO3	**10**
		OR			
	b.	Demonstrate the Tabular Dyna-Q algorithm.	L3	CO5	**10**
13.	**a.**	Describe the Q-learning algorithm. Examine the relationship between Q-learning and the Bellman optimality equations.	L3	CO4	**10**
		OR			
	b.	Give your judgement on the relative performance between the on-policy nature of SARSA and the off-policy nature of Q-learning.	L5	CO6	**10**